Rolling-burden alarm and temporal granularity

1. What the alarm is

The alarm is a **prospective rolling-burden** rule on behavioural topic probabilities from LDA / DCABP.

At each decision time (t):

1. Build **pUF** = P(topic 3) + P(topic 4) + P(topic 5) (unfavourable day-types).
2. Aggregate pUF over a **lookback window W** into a **burden** score.
3. Raise an alarm if burden ≥ threshold (\theta).
4. Label the decision as positive if progression (PD) occurs within a future **horizon H** after (t).

Only decisions with enough history (full W) and enough follow-up to score the label (or a known PD time) are kept. Those eligible ((patient, t)) pairs are the **decision points**.

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2. Parameters that can be varied

Parameter	Role	Typical choices
W (lookback)	How much past burden is summarised at (t)	60 / 90 / 120 days (≈ 2 / 3 / 4 months)
H (horizon)	How far ahead PD counts as a hit	90 / 120 / 180 days (≈ 3 / 4 / 6 months)
Granularity	How often decisions are made and how LDA is built	daily, weekly, biweekly, monthly
Score family	How pUF / burden are constructed	**sampled** vs **collapsed** (below)
θ (threshold)	Alarm cutoff on the burden scale	Fixed operating point (θ = 1.60 on monthly *sum* scale), or Youden for other constructions
Event / censor time	Clock used to stop landmarks and label PD	user clock BBDD (`fecha_fin − fecha_inicio`)

Sampled vs collapsed

Family	Idea
Sampled	

Family	Idea
	Daily sliding LDA (30-day window) → daily pUF; weekly / biweekly / monthly only change *when* you decide (same underlying daily signal). Burden = rolling mean over the last **W days**.
Collapsed	One LDA per non-overlapping block (7 / 14 / 30 d) → block pUF; burden = rolling mean over the last `round(W/block)` blocks.

Collapsed block pUF is **not** the mean of daily pUF inside the block: each family is its own LDA construction.

Monthly collapsed landmarks

On the monthly scale, month (m) is placed at day (t = m \times 30). A landmark is kept only if (t) is strictly before the event/censor time. Burden is the **sum** of pUF over the last W months (so thresholds are on a sum scale, θ = 1.60, not the mean scale ~0.53 used elsewhere).

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3. Experiment in this report

We fix a main operating region and then sweep the knobs above:

1. **Monthly collapsed baseline** — W = 3 months, H = 4 months, user clock BBDD (`t_evento = fecha_fin − fecha_inicio`) as event/censor clock, fixed θ = 1.60.
2. **Granularity comparison** — W = 90 d, H = 120 d; sampled (daily→monthly) vs collapsed (weekly / biweekly / monthly).
3. **W × H grid** — W ∈ {60, 90, 120} d × H ∈ {90, 120, 180} d.
4. **Individual trajectories** — example patients with and without PD.

Outputs live under `results/alarm/` (`tables/` , `figures/` , `analysis_figures/` , `runs/Evento_DB1_clock/`). Primary monthly landmarks: `runs/Evento_DB1_clock/paper_W3_H4/decision_points_W3_H4.csv` (**clock BBDD**, n=1016).

Delivery summary figures (high-res, clock-aligned, AUROC with patient-bootstrap CIs): `analysis_figures/` .

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4. Results — monthly collapsed baseline (W = 3 mo, H = 4 mo)

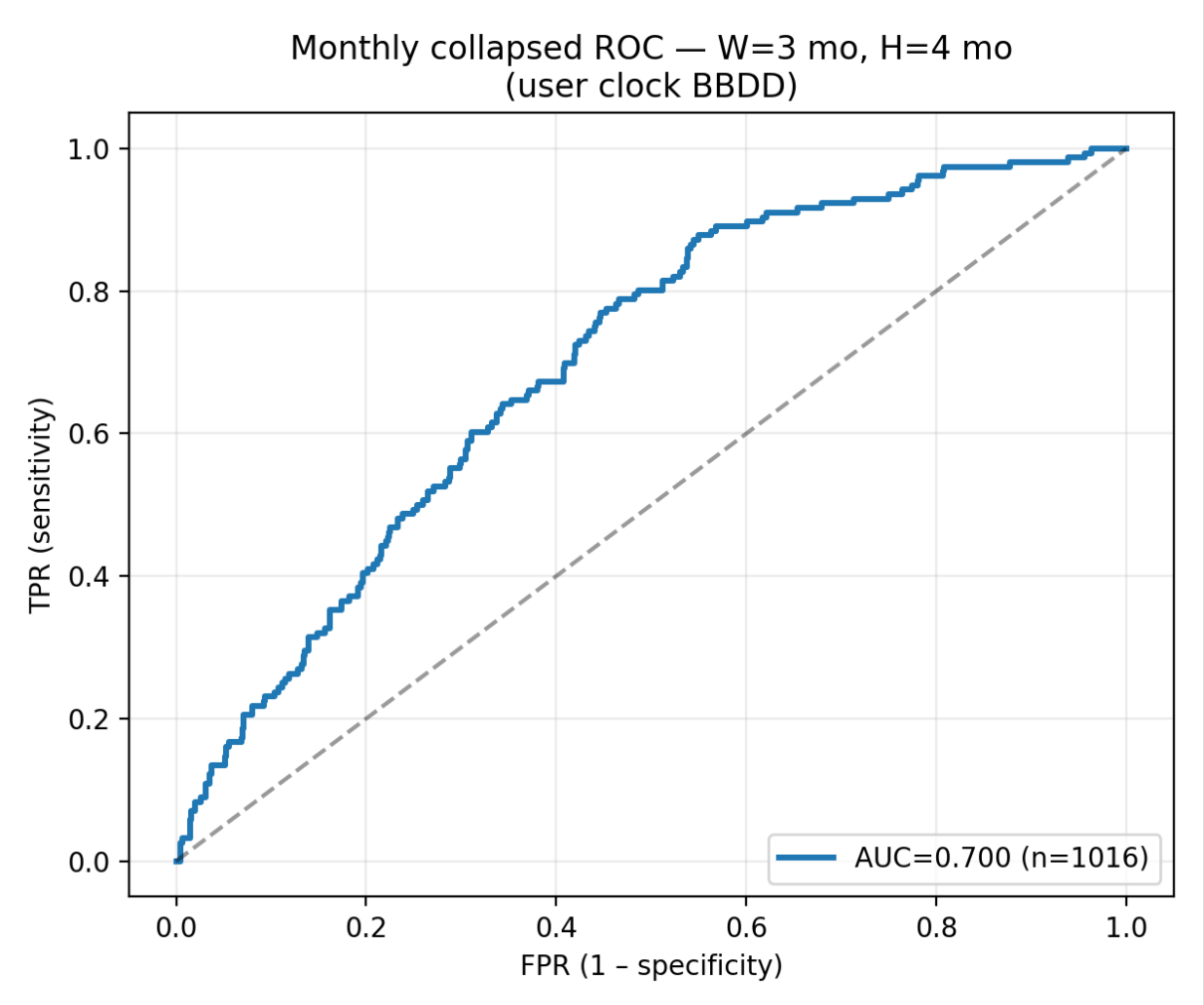
Score = **sum** of pUF over W months; cutoff: `month × 30 < t_evento` (clock BBDD).
n = **1016** decision points · **116** patients · **156** positives.

Metric	Value
ROC-AUC	**0.700**
Sens / Spec (θ≈1.60, sum scale)	0.673 / 0.616
PPV / NPV	0.241 / 0.912
TP / FN / FP / TN	105 / 51 / 330 / 530

Threshold scale (same rule, two writings):

Scale	θ	Relation
Sum (as used here)	**1.60**	`sum` of last W=3 monthly pUF
Mean (equivalent)	**≈ 0.533**	`1.60 / 3`

Operating point used throughout (baseline metrics + trajectories): **θ = 1.60** (sum scale). AUROC describes the whole ROC and does **not** pick a threshold. A pure Youden cut on this clock (~1.06) over-alarms (sens≈0.88 / spec≈0.45) and is **not** used. Final clinical θ to be confirmed.



5. Results — granularity (W = 90 d, H = 120 d)

Monthly collapsed row: manuscript θ=1.60 (same as §4). Other rows: construction-wise Youden on their own score scale (for a single comparable operating point per row — not the clinical cut-off of the paper).

Clinical clock: all rows use `clock_bbdd.xlsx` (`t_evento = fecha_fin − fecha_inicio`). **AUROC uncertainty:** 95% CIs from **patient-clustered bootstrap** (2000 reps). See `analysis_figures/auc_ci_W90_H120.csv`.

Granularity	n	AUC	95% CI	Sens	Spec	PPV	NPV
daily (sampled)	36495	0.683	0.610–0.749	0.845	0.437	0.207	0.942
weekly sampled	5262	0.682	0.613–0.753	0.845	0.437	0.206	0.942
biweekly sampled	2664	0.678	0.608–0.749	0.824	0.456	0.205	0.939
monthly sampled	1283	0.679	0.612–0.749	0.816	0.475	0.208	0.939
weekly collapsed	4137	0.700	0.620–0.773	0.734	0.590	0.242	0.925
biweekly collapsed	2130	0.697	0.620–0.769	0.781	0.537	0.245	0.928
monthly collapsed (30 d)	**1016**	**0.700**	0.623–0.770	0.673	0.616	0.241	0.912

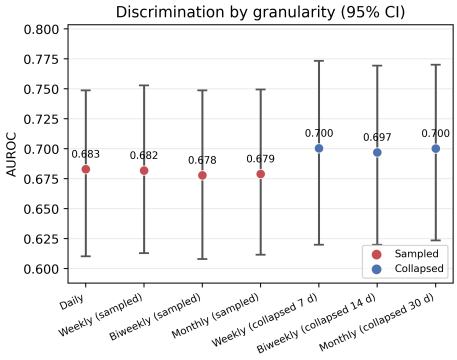
Short read

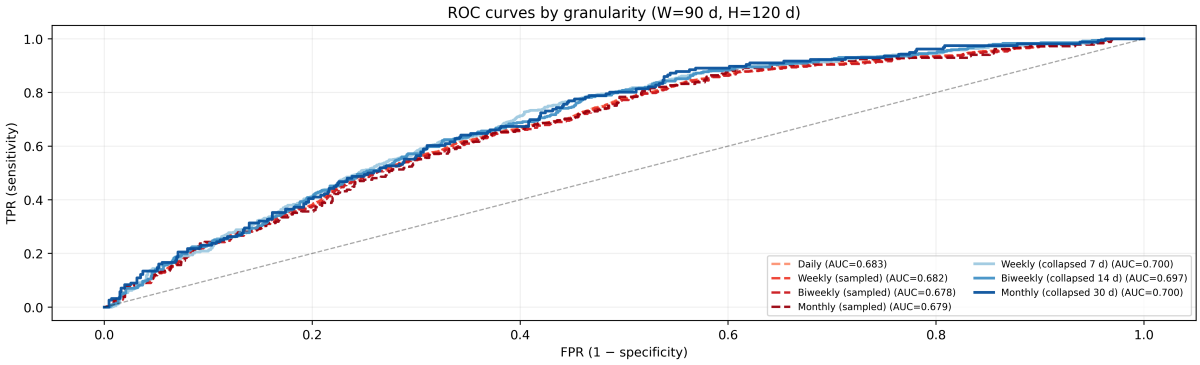
- **Sampled** (daily→monthly): AUC nearly flat (~0.680). Subsampling does not improve discrimination; it only reduces n.
- **Collapsed** improves somewhat vs sampled (~0.699). Highest point estimate here: **weekly collapsed (AUC 0.700)**; CIs overlap across collapsed configs.
- PPV remains low (~0.205–0.245); NPV high. High-resolution delivery panel (metrics table + AUROC points with 95% CI + ROC; no winner star):

Alarm performance across temporal granularities

Metrics at operating θ (monthly θ=1.60; else Youden; W=90 d, H=120 d)

Granularity	N	AUROC	Sens	Spec	PPV	NPV
Daily	36495	0.683	0.845	0.437	0.207	0.942
Weekly · sampled	5262	0.682	0.845	0.437	0.206	0.942
Biweekly · sampled	2664	0.678	0.824	0.456	0.205	0.939
Monthly · sampled	1283	0.679	0.816	0.475	0.208	0.939
Weekly · collapsed	4137	0.700	0.734	0.590	0.242	0.925
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Monthly · collapsed	1016	0.700	0.673	0.616	0.241	0.912





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6. Results — W × H grid

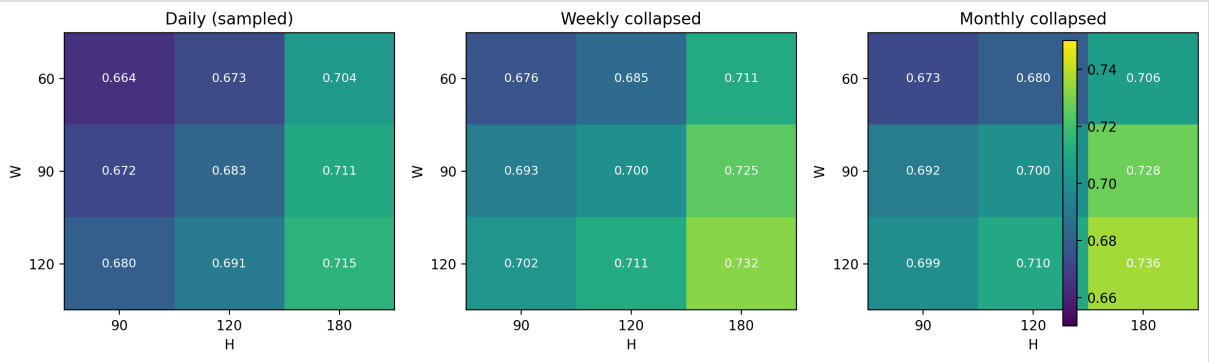
Grid: **W ∈ {60, 90, 120} d** × **H ∈ {90, 120, 180} d**.

AUC (selected)

	H=90	H=120	H=180
Daily W=60/90/120	0.664 / 0.672 / 0.680	0.673 / 0.683 / 0.691	0.704 / 0.711 / 0.715
Weekly collapsed	0.676 / 0.693 / 0.702	0.685 / 0.700 / 0.711	0.711 / 0.725 / 0.732
Monthly collapsed	0.673 / 0.692 / 0.699	0.680 / 0.700 / 0.710	0.706 / 0.728 / 0.736

Read

- ↑**H** → AUC rises monotonically.
- ↑**W** → AUC rises moderately (more historical load).
- Ranking **collapsed ≳ sampled** holds across almost the whole grid.
- W≈90 / H≈120 sits in the mid region of the map.



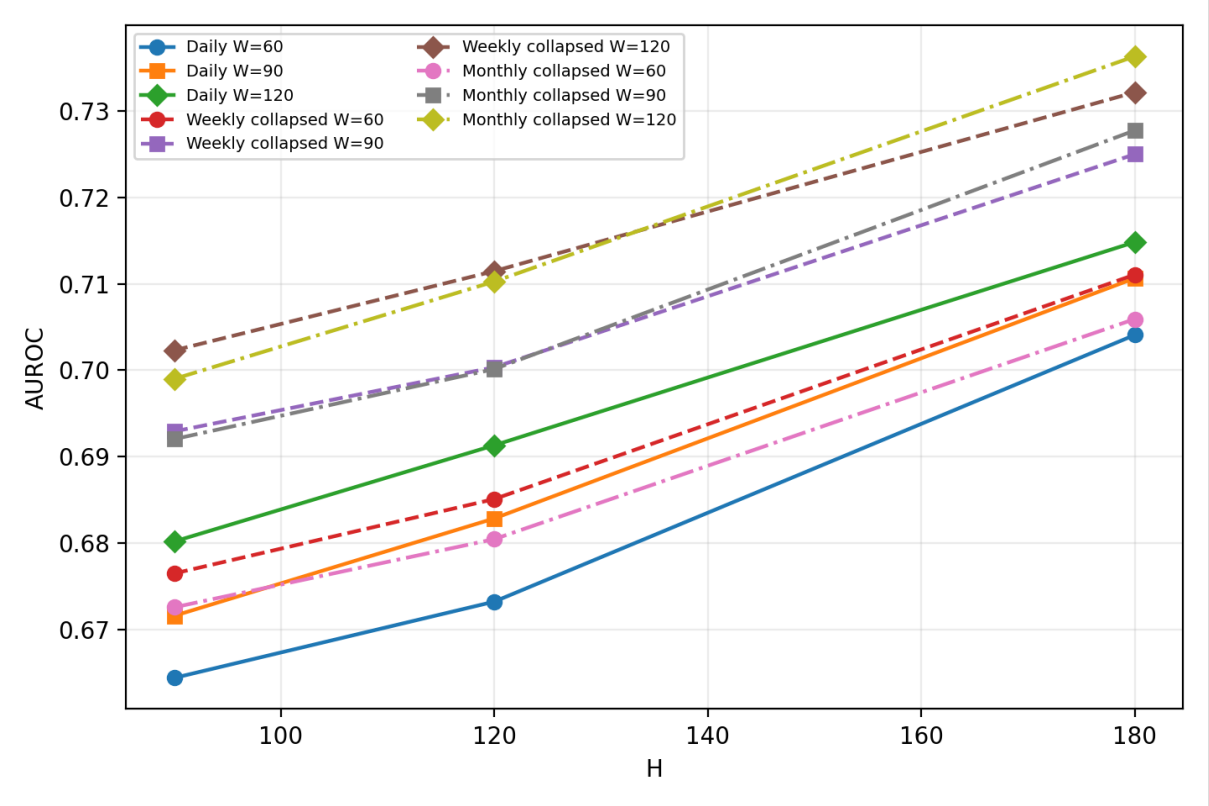


Table: `runs/Evento_DB1_clock/granularity/granularity_sweep_summary.csv`

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7. Results — individual trajectories

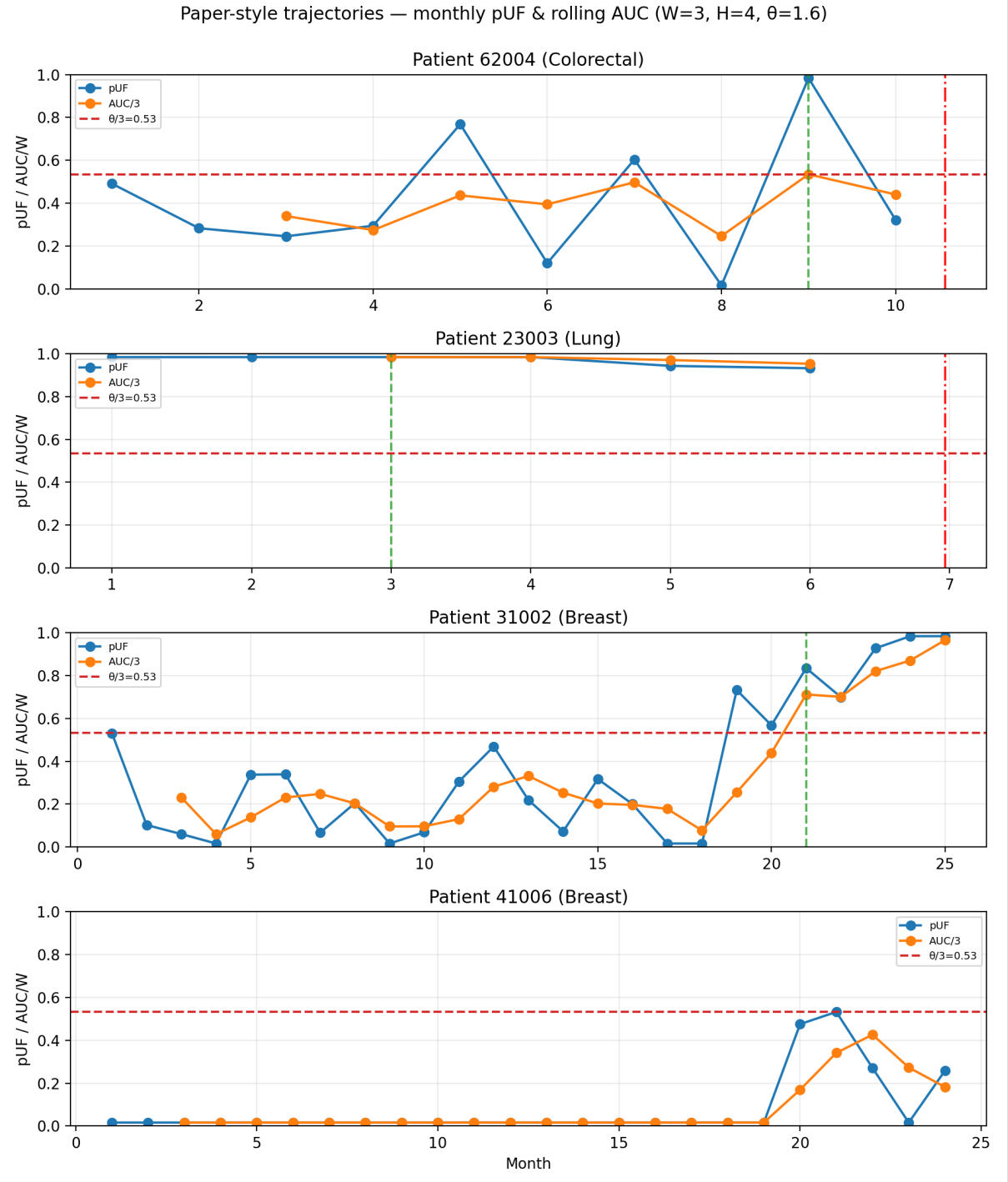
Patients: **62004**, **23003**, **31002**, **41006**.

ID	Profile	Notes
62004	PD (day 317)	1st alarm ~month 9 (lead before PD)
23003	PD (day 209)	1st alarm ~month 3 (lead before PD)
31002	No PD	1st alarm ~month 21 with no event in follow-up

ID	Profile	Notes
41006	No PD	Below threshold; no alarm

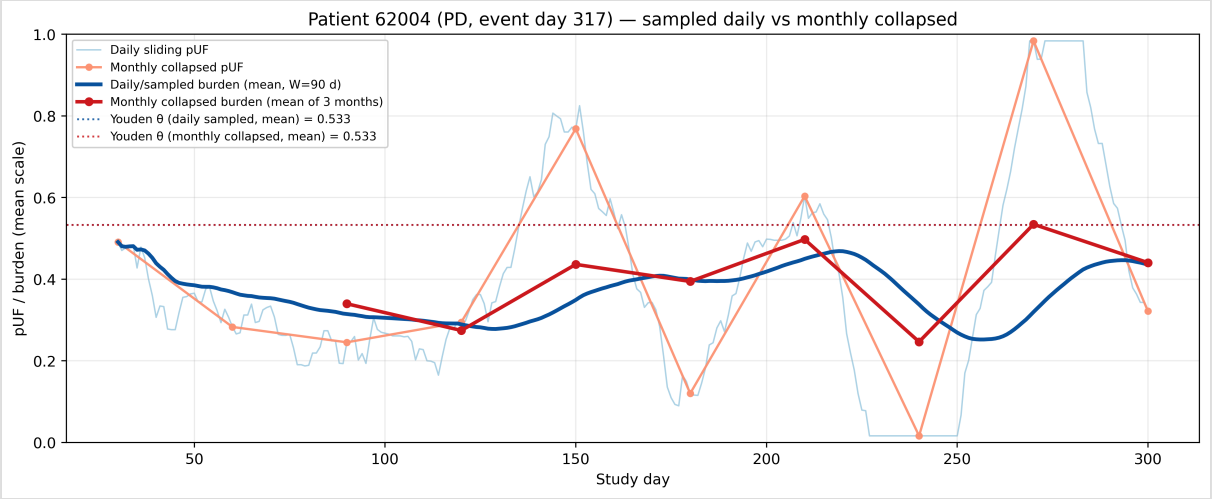
7.1 Monthly collapsed (pUF + rolling burden)

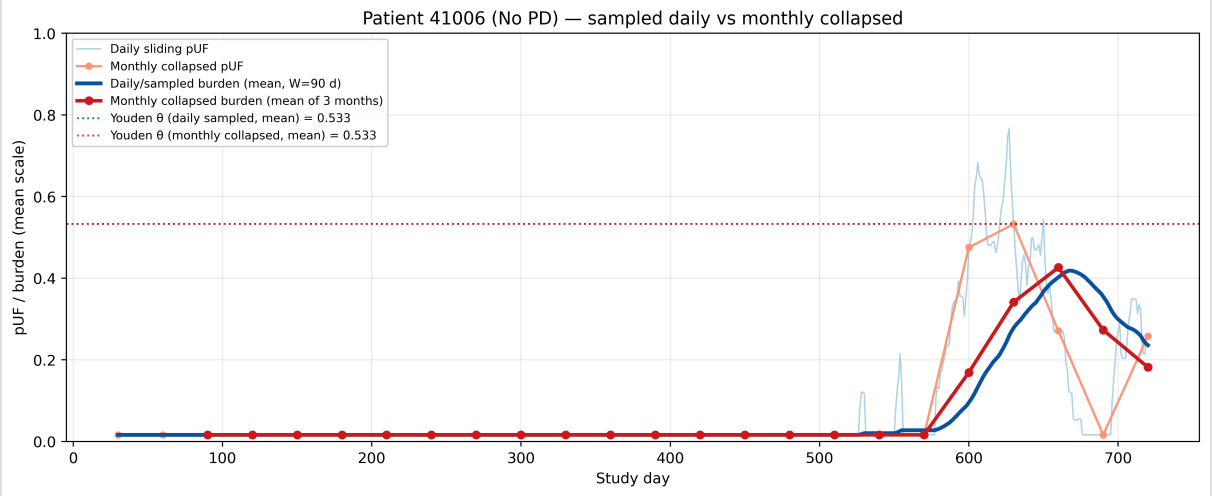
W=3 months, H=4 months, θ=1.60 (sum scale; mean-scale θ/W≈0.533).



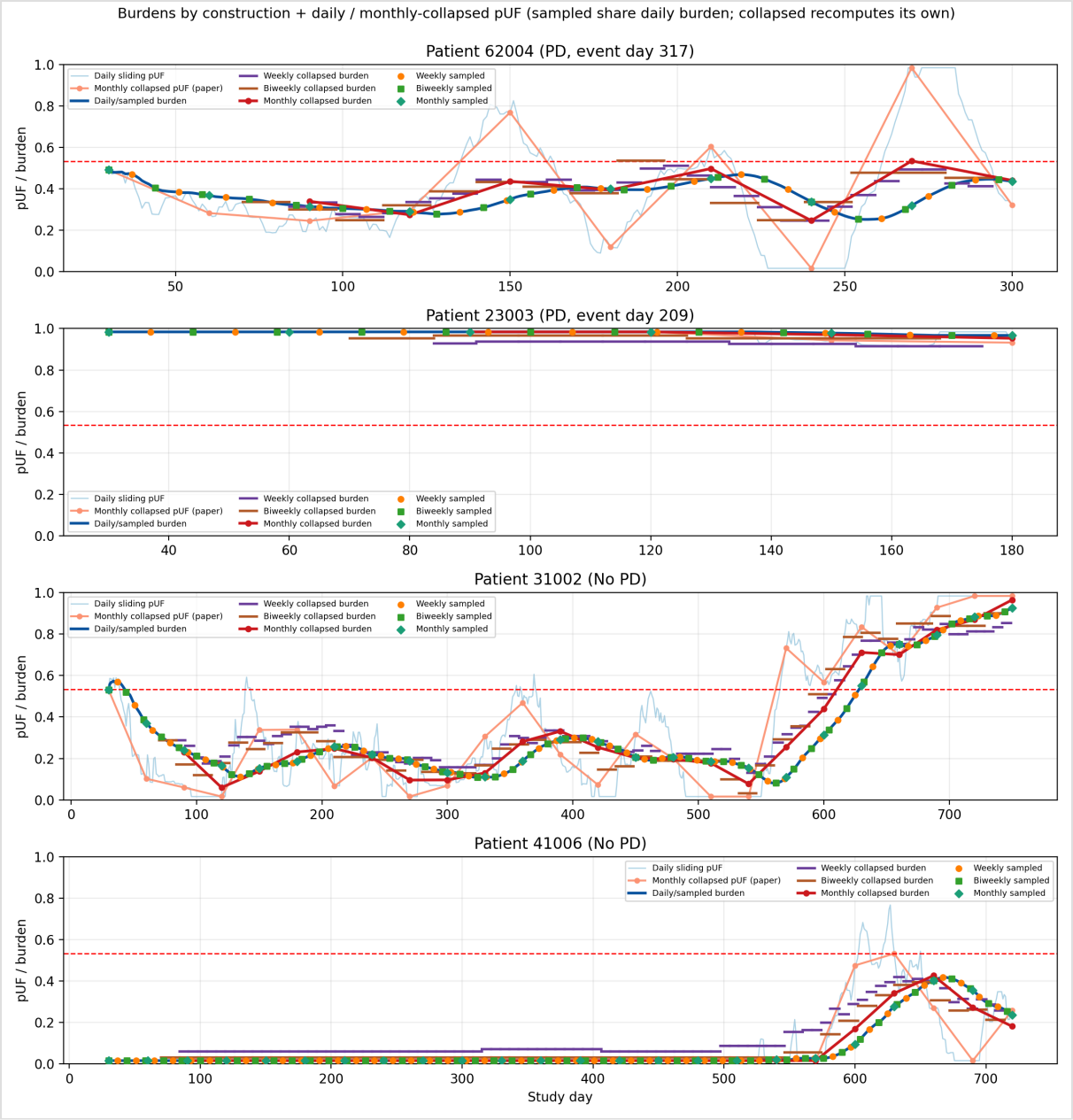
7.2 Sampled vs collapsed (W = 90 d)

Simplified high-res delivery plots (daily/sampled burden vs monthly collapsed; clock BBDD; manuscript-aligned θ labelled):

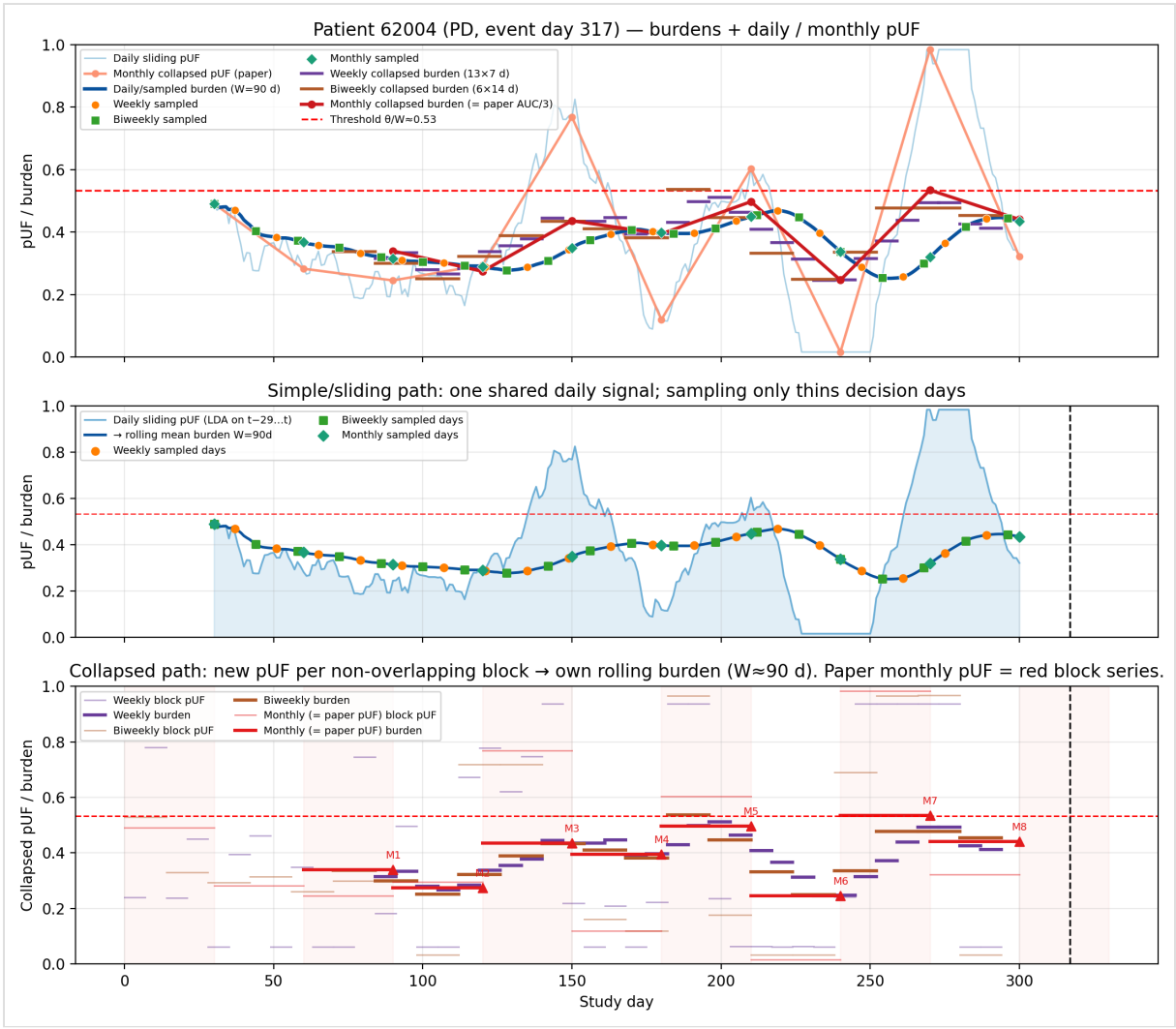


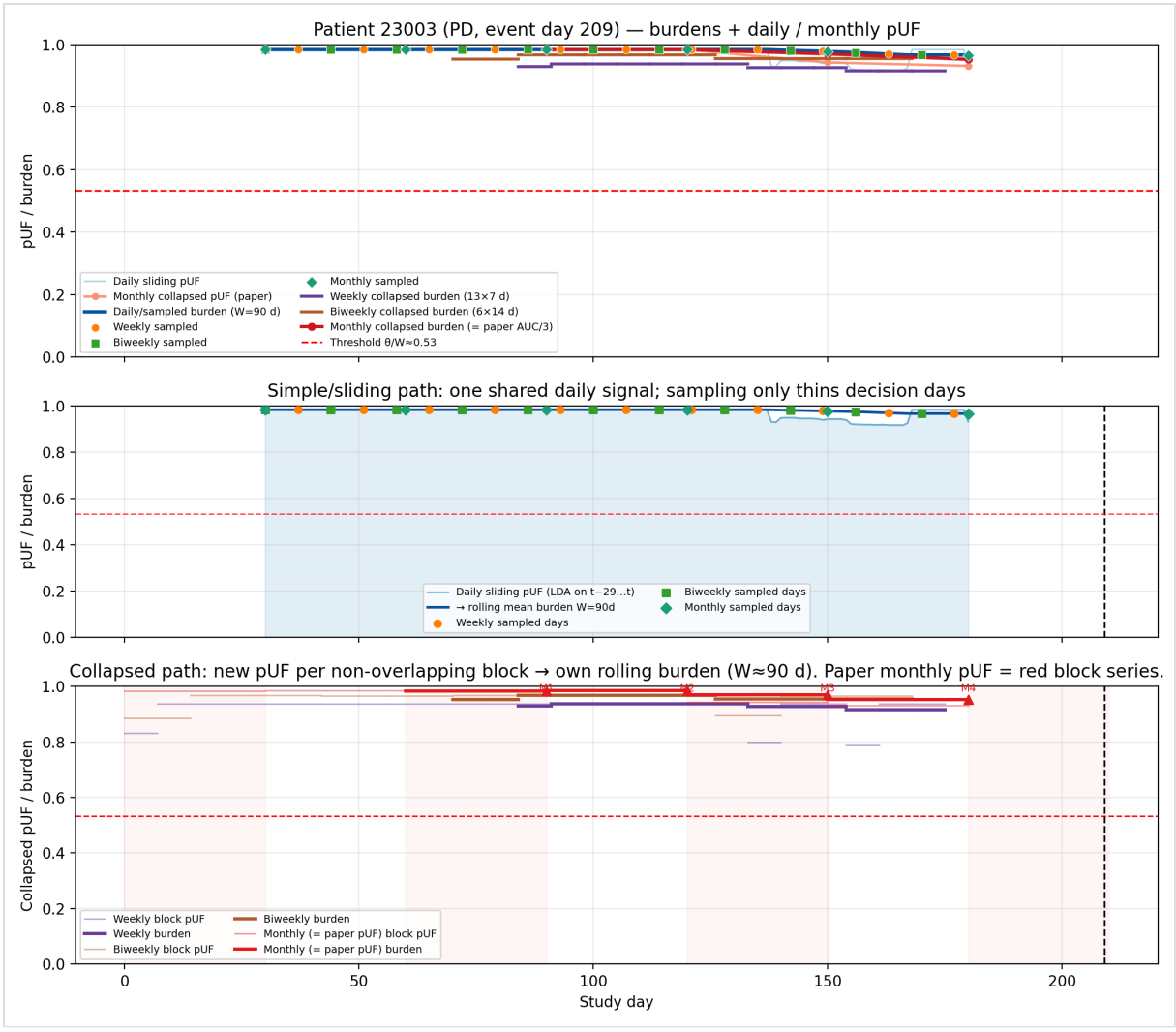


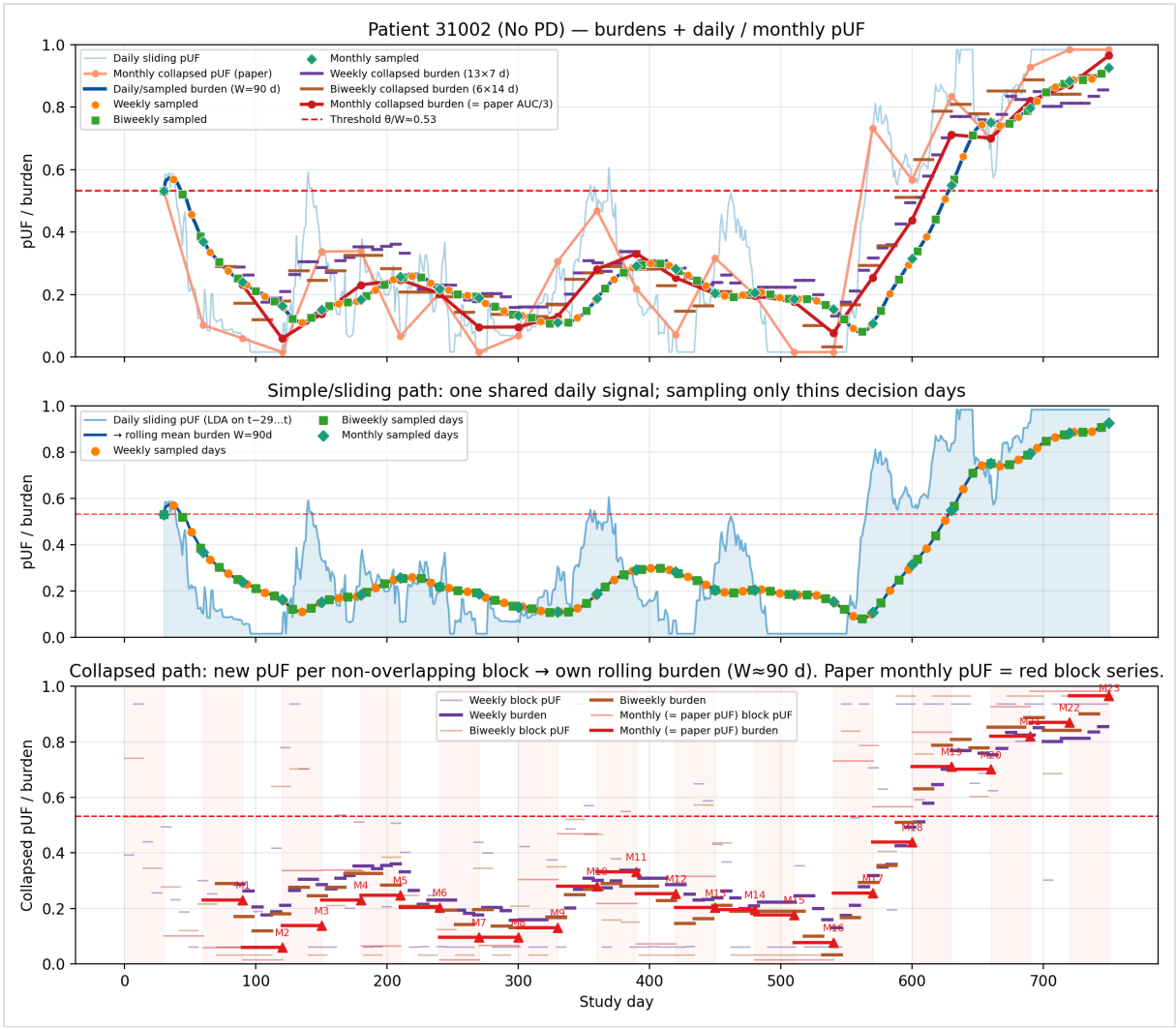
Full multi-panel trajectories:

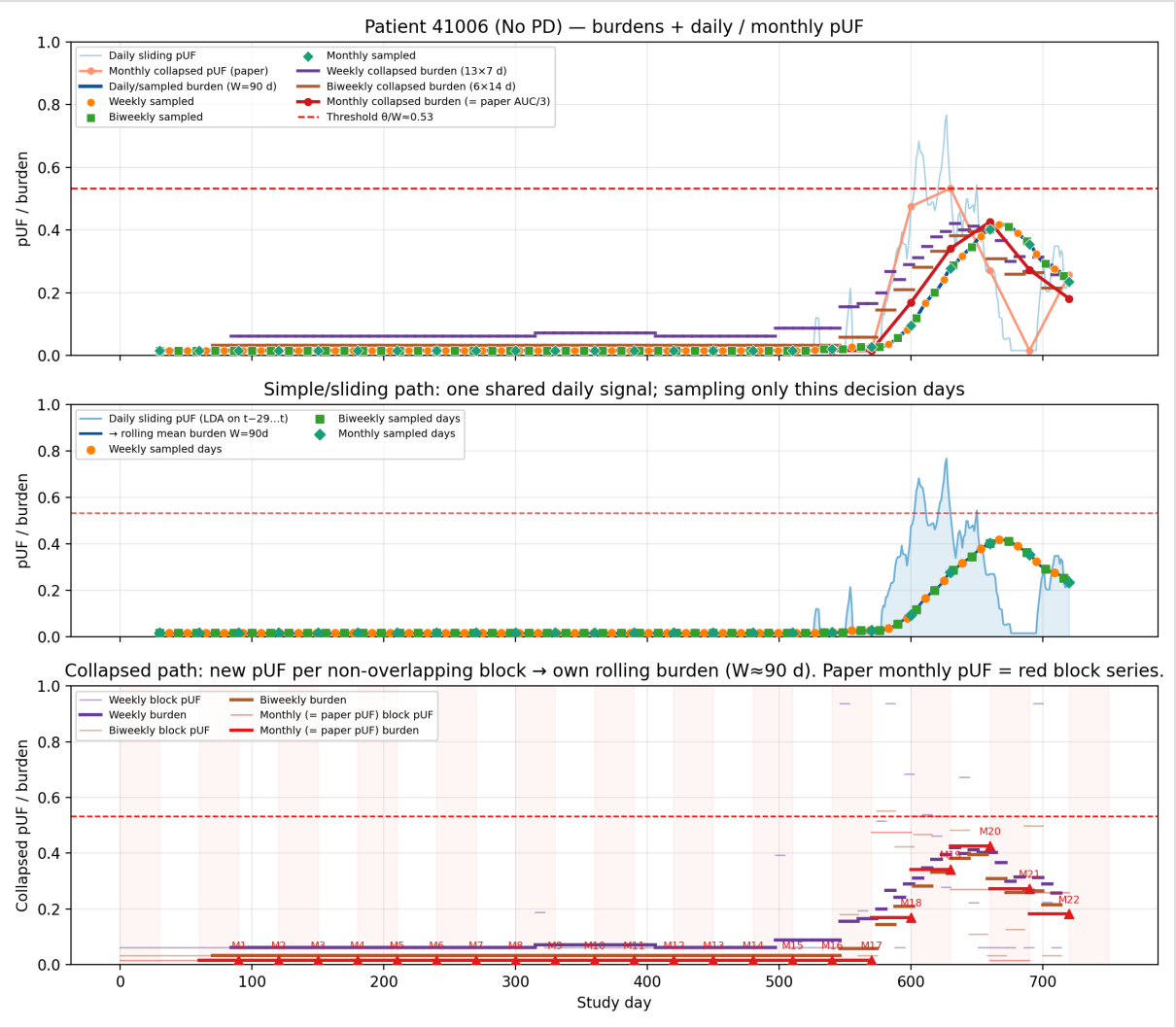


- **Blues:** daily sliding pUF + daily/sampled burden
- **Reds:** monthly collapsed pUF + monthly collapsed burden
- Steps: weekly / biweekly collapsed burden
- Markers: sampled evaluation (7 / 14 / 30 d) on the daily burden


Patient 62004 (PD, event day 317) — burdens + daily / monthly pUF
Daily sliding pUF
Monthly collapsed pUF (paper)
Daily/sampled burden (W=90 d)
Weekly sampled
Biweekly sampled
Monthly sampled
Weekly collapsed burden (13×7 d)
Biweekly collapsed burden (6×14 d)
Monthly collapsed burden (= paper AUC/3)
Threshold θ/W=0.53
pUF / burden
Simple/sliding path: one shared daily signal; sampling only thins decision days
Daily sliding pUF (LDA on t−29…t)
→ rolling mean burden W=90d
Weekly sampled days
Biweekly sampled days
Monthly sampled days
pUF / burden
Collapsed path: new pUF per non-overlapping block → own rolling burden (W≈90 d). Paper monthly pUF = red block series.
Weekly block pUF
Weekly burden
Biweekly block pUF
Biweekly burden
Monthly (= paper pUF) block pUF
Monthly (= paper pUF) burden
M1
M3
M4
M5
M6
M7
M8
Collapsed pUF / burden
Study day


Patient 23003 (PD, event day 209) — burdens + daily / monthly pUF
pUF / burden
Daily sliding pUF
Monthly collapsed pUF (paper)
Daily/sampled burden (W=90 d)
Weekly sampled
Biweekly sampled
Monthly sampled
Weekly collapsed burden (13×7 d)
Biweekly collapsed burden (6×14 d)
Monthly collapsed burden (= paper AUC/3)
Threshold θ/W≈0.53
Simple/sliding path: one shared daily signal; sampling only thins decision days
pUF / burden
Daily sliding pUF (LDA on t−29…t)
→ rolling mean burden W=90d
Weekly sampled days
Biweekly sampled days
Monthly sampled days
Collapsed path: new pUF per non-overlapping block → own rolling burden (W≈90 d). Paper monthly pUF = red block series.
Collapsed pUF / burden
M1
M2
M3
M4
Weekly block pUF
Weekly burden
Biweekly block pUF
Biweekly burden
Monthly (= paper pUF) block pUF
Monthly (= paper pUF) burden
Study day


Patient 31002 (No PD) — burdens + daily / monthly pUF
Daily sliding pUF
Monthly collapsed pUF (paper)
Daily/sampled burden (W=90 d)
Weekly sampled
Biweekly sampled
Monthly sampled
Weekly collapsed burden (13×7 d)
Biweekly collapsed burden (6×14 d)
Monthly collapsed burden (= paper AUC/3)
Threshold θ/W=0.53
pUF / burden
Simple/sliding path: one shared daily signal; sampling only thins decision days
Daily sliding pUF (LDA on t−29…t)
→ rolling mean burden W=90d
Weekly sampled days
Biweekly sampled days
Monthly sampled days
pUF / burden
Collapsed path: new pUF per non-overlapping block → own rolling burden (W≈90 d). Paper monthly pUF = red block series.
Weekly block pUF
Weekly burden
Biweekly block pUF
Biweekly burden
Monthly (= paper pUF) block pUF
Monthly (= paper pUF) burden
Collapsed pUF / burden
Study day



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8. How to reproduce

```
 # Paper baseline + granularity (user clock BBDD)
python scripts/03_analysis/alarm/run_alarm_pipeline.py \
  --clock data/revision/clock_bbdd.xlsx --run-id Evento_DB1_clock \
  --with-granularity --skip-lda

# Monthly / granularity trajectories (θ=1.60)
python scripts/03_analysis/alarm/plot_patient_trajectories.py \
  --patients 62004,23003,31002,41006 --clock data/revision/clock_bbdd.xlsx
python scripts/03_analysis/alarm/plot_granularity_trajectories.py \
  --patients 62004,23003,31002,41006 --W 90 --clock data/revision/clock_bbdd.xlsx

# High-res delivery figures (clock-aligned; AUROC patient-bootstrap CIs)
python scripts/03_analysis/alarm/plot_analysis_figures.py \
  --clock data/revision/clock_bbdd.xlsx --n-boot 2000
```

```
# This report (main-branch markdown layout)
python scripts/03_analysis/alarm/write_granularity_report.py --run-id Evento_DB1_clock
```

File map: `README.md`. Delivery panel: `analysis_figures/`.

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9. Conclusions

1. **Subsampling** the same daily signal (sampled) **does not improve** discrimination: AUC stays ~**0.680** from daily→monthly.
2. **Collapsing** LDA into non-overlapping blocks (7/14/30 d) **does improve** somewhat: AUC ~**0.699** at W=90 d, H=120 d.
3. The **highest point estimate** in that setting is **weekly collapsed** (AUC **0.700**); patient-bootstrap 95% CIs overlap across collapsed configs.
4. ↑**W** and ↑**H** raise AUC in an ordered way; ranking collapsed ≳ sampled holds on the grid.
5. **PPV** stays low (~**0.222**): the alarm discriminates but yields many false positives at useful thresholds.
6. Individually, the score can anticipate PD (lead time), mark persistently high burden, or fire late alarms with no event — useful for interpretation beyond aggregate metrics.
7. The delivery summary figure uses the **same clinical follow-up field (clock BBDD)** for all granularities as the monthly baseline (n=1016), with **AUROC 95% CIs from patient-level bootstrap**.

Generated 2026-10-21 · run `Evento_DB1_clock` *· clock* `data/revision/clock_bbdd.xlsx`

Generated 2026-10-21